### 1. Item 2: Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem

**(Judge whether a mathematical result is reasonable within the problem's real-world context)**

- Highly relevant: Students must connect abstract numerical data (year, price, mileage) to realistic car-buying criteria and determine which vehicles are "acceptable" or "unacceptable" based on those criteria.
- For example, although Castel has the lowest price (4,250 zeds), its mileage is 128,000 km—exceeding the 120,000 km limit. Even if other conditions are met, this option is "unreasonable" in the given context and must be eliminated.
- This reflects the ability to judge whether a result "makes sense" in reality, rather than merely performing numerical comparisons.

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### 2. Item 9: Understand the relationship between the context of the problem and the representation of the mathematical solution...

**(Understand how the real-world context relates to the mathematical representation, and use this understanding to interpret the feasibility and limitations of the solution)**

- Highly relevant: The problem presents data in a table (a structured mathematical representation), and students must map this back onto the real-world decision-making scenario of purchasing a car.
- Each column (year, price, mileage) is not just an isolated number but represents a key practical factor in evaluating a vehicle's desirability (e.g., "newer year is generally better," "lower mileage often means better condition").
- Selecting Bolte as the answer involves not only logical filtering but also an implicit judgment about what constitutes a "feasible" real-world choice.

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### 3. Item 1: Interpret a mathematical result back into the real-world context

**(Translate a mathematical conclusion back into meaningful real-world terms)**

- Moderately relevant: After completing the filtering process, students must interpret the mathematical conclusion—"Bolte is the only car meeting all criteria"—as the real-world recommendation: "Chris should buy the Bolte."
- Although the question doesn't explicitly ask for a recommendation, selecting the correct answer inherently involves this act of contextual interpretation.

### 4. Item 3: Identify the limits of the model used to solve a problem

**(Recognize the limitations of the model applied)**

- Low but present relevance: The model provided is highly simplified (only three criteria). In reality, car buyers would also consider factors like mechanical condition, service history, fuel efficiency, and insurance costs.
- While the task doesn't require students to point out these omissions, a critically minded student might recognize: "Choosing a car based only on these three factors may not be sufficient."
- Thus, the question indirectly touches on awareness of model limitations, though it is not a primary assessment focus.

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### 5. Item 6: Use knowledge of how the real world impacts outcomes... to make contextual judgments

**(Apply real-world knowledge to adjust or apply results appropriately)**

- Weakly relevant: The problem provides clear, unambiguous constraints with no room for negotiation or interpretation (e.g., no scenarios like "price slightly over budget but negotiable" that would require real-life experience).
- Therefore, this competency plays a minimal role in this particular task.

### 1. 第2条：Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem

**（判断数学结果在问题情境中是否合理）**

- 高度相关：学生需要将抽象的数值（年份、价格、里程数）与现实购车需求对应起来，判断哪些车“符合”或“不符合”条件。
- 例如，Castel 虽然价格最低（4250 zeds），但其行驶里程为128,000 km，超过了120,000 km的限制——即使其他条件满足，该结果在上下文中“不合理”，应被排除。
- 这体现了对结果是否“在现实中成立”的判断，而非仅做数值比较。

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### 2. 第9条：Understand the relationship between the context of the problem and representation of the mathematical solution...

**（理解问题情境与数学表示之间的关系，并据此解释解的可行性与局限）**

- 高度相关：题目以表格形式呈现数据（数学/结构化表示），学生需将其映射回“买车”这一真实决策场景。
- 每一列（年份、价格、里程）都不是孤立数字，而是代表车辆状况的关键现实指标。学生必须理解这些变量在购车语境下的意义（如“年份越新通常越好”“里程越低车况可能越好”）。
- 最终选择 Bolte 不仅是逻辑筛选的结果，也隐含了对“可行解”的现实判断。

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### 3. 第1条：Interpret a mathematical result back into the real world context

**（将数学结果还原到现实情境中）**

- 中等相关：学生完成筛选后，需将“Bolte 是唯一满足所有条件的车”这一数学结论，转化为“Chris 应该买 Bolte”这一现实建议。
- 虽然题目未明确要求提出建议，但选择正确答案本身就隐含了这种解释过程。

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### 4. 第3条：Identify the limits of the model used to solve a problem

**（识别所用模型的局限性）**

- 较低但存在相关性：题目提供的模型非常简化（仅三个指标），现实中购车还需考虑车况、维修记录、油耗、保险等。
- 虽然题目未要求学生指出这些缺失因素，但有批判性思维的学生可能会意识到：“仅凭这三个条件选车可能不够全面”。
- 因此，该题间接触及模型局限性的意识，但未作为考查重点。

### 5. 第6条：Use knowledge of how the real world impacts the outcomes... to make contextual judgments

**（利用现实知识判断如何调整或应用结果）**

- 弱相关：本题条件明确、无模糊地带，不需要对结果进行“调整”。例如，没有出现“价格略超但可议价”之类需现实经验介入的情形。
- 因此，该能力在此题中体现较弱。